

From Traditional Semantic Analysis to Generative AI: Research on Cybersecurity and Privacy Challenges in Cloud-Based Educational task Automation

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摘要

大語言模型 (LLM) 的快速發展已將對話代理轉變為自動化複雜資訊任務的強大工具。然而，在教育 and 行政環境中部署基於 LLM 的系統往往會受到後端複雜性和高維護需求的阻礙。本研究介紹 GenAI Cloud Executive Secretary，這是一個用於教育任務自動化的輕量級、無伺服器對話框架。該框架基於 Google Apps Script (GAS) 構建，以 Gemini 2.0 Flash 作為推理核心，並與 LINE Messaging API 集成，提供自然語言互動，並透過意圖識別和結構化回饋自主協調 Google Workspace 服務（例如 Google 雲端硬碟、Google 日曆、Gmail 等）。

除了上述功能之外，本研究所提出框架更強調雲端安全和隱私，透過應用安全設計原則來降低未經授權的存取、資料外洩和對抗性提示等風險。細粒度存取控制、加密通訊和權限管理等機制確保符合教育資料保護標準。比較分析凸顯了基於規則的自動化和 LLM 驅動的方法在面臨錯誤分類、幻覺和惡意查詢操縱等威脅時的不同復原能力。

在教育環境中的案例實作證明，我們提出架構不僅提升了任務效率和使用者的體驗，還能以低成本和最小的技術門檻提供安全、隱私保護的自動化功能。研究結果凸顯了我們所提框架在可擴展教育任務自動化方面的潛力，以及其對雲端安全、應用服務安全和隱私感知型人工智慧應用的廣泛影響。

關鍵詞：大語言模型、教育任務自動化、無伺服器對話框架、雲端安全與隱私、應用服務安全

ABSTRACT

The rapid growth of large language models (LLMs) has advanced conversational agents into powerful tools for automating complex information tasks. However, deploying LLM-based systems in educational and administrative contexts is often hindered by backend complexity and high maintenance demands. This study introduces the GenAI Cloud Executive Secretary, a lightweight, serverless conversational framework for educational

task automation. Built on Google Apps Script (GAS) with Gemini 2.0 Flash as the reasoning core, the framework integrates with the LINE Messaging API to provide natural language interaction and autonomously coordinates Google Workspace services (e.g., Drive, Calendar, Gmail) through intent recognition and structured responses.

Beyond functionality, the framework emphasizes cloud security and privacy by applying security-by-design principles to mitigate risks such as unauthorized access, data leakage, and adversarial prompts. Mechanisms including fine-grained access control, encrypted communication, and permission management ensure compliance with educational data protection standards. Comparative analysis highlights the differing resilience of rule-based automation and LLM-driven approaches when exposed to threats like misclassification, hallucination, and malicious query manipulation.

Case implementation demonstrates that our proposed architecture not only improves task efficiency and user experience but also provides secure, privacy-preserving automation at low cost with minimal technical barriers. The findings underscore the framework's potential for scalable educational task automation and its broader implications for cloud security, application service safety, and privacy-aware AI adoption.

Keywords: Large Language Models (LLMs), Educational Task Automation, Serverless Conversational Framework, Cloud Security and Privacy, Application Service Safety

I. INTRODUCTION

The rapid advancement of generative artificial intelligence (AI) has profoundly transformed the landscape of educational task automation, shifting from traditional semantic analysis methods to sophisticated AI-driven approaches [1]. This evolution brings unprecedented opportunities to enhance learning experiences and streamline educational processes by leveraging cloud-based platforms. However, the integration of generative AI within these cloud environments also raises significant cybersecurity and privacy challenges, which merit rigorous investigation. This paper explores the emerging threats and vulnerabilities associated with deploying generative AI technologies in educational settings, focusing on the protection

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of sensitive data and the integrity of automated systems. By critically examining current security frameworks and identifying gaps, this research aims to contribute to the development of robust strategies that safeguard both learners and educators in the increasingly digital and AI-powered educational ecosystem [2].

As educational institutions progressively adopt cloud-based solutions to facilitate task automation, the reliance on generative AI models introduces complex security concerns [3]. These concerns stem from the models' inherent capabilities to generate and process vast amounts of data, which often include personally identifiable information and academic records. Traditional semantic analysis techniques, while effective in structured and rule-based environments, struggle to address the dynamic and evolving nature of AI-generated content, thus necessitating new approaches to cybersecurity and privacy protection [4].

Moreover, the cloud infrastructure supporting these AI systems presents additional attack surfaces, including vulnerabilities related to data transmission, storage, and multi-tenant access. The increasing sophistication of cyber-attacks targeting educational technologies underscores the urgency for proactive research and comprehensive security frameworks tailored to this context. This study systematically investigates the current state of cybersecurity measures in cloud-based educational automation, analyzing the interplay between AI capabilities and potential threats [5].

By integrating insights from both traditional semantic analysis and modern generative AI methodologies, this research outlines challenges and proposes novel solutions to mitigate risks. Ultimately, it seeks to foster a secure and privacy-respecting environment that can fully leverage the benefits of AI-driven automation in education while minimizing adverse impacts on stakeholders [6].

This paper is structured as follows: First, we provide a comprehensive review of traditional semantic analysis techniques and their applications in educational task automation, highlighting their limitations in the current AI-driven landscape. Next, we examine the transformative role of generative AI technologies, focusing on their architecture, functionalities, and integration within cloud-based educational platforms. Subsequently, we delve into the core cybersecurity and privacy challenges arising from this integration, including data breaches, unauthorized access, and algorithmic vulnerabilities.

Following the identification of these challenges, we discuss existing defense mechanisms and their adequacy in addressing the unique risks posed by generative AI in education. To bridge identified gaps, we propose an enhanced security framework that combines advanced encryption methods, privacy-preserving AI techniques, and adaptive threat detection systems. This framework aims to ensure data confidentiality, integrity, and availability while maintaining the operational efficiency of automated educational tasks.

Finally, this study concludes with future research directions and practical recommendations for policymakers, educators, and technology developers. Through this interdisciplinary approach, the study contributes to a deeper understanding of securing

cloud-based educational automation in the era of generative AI, promoting safer and more trustworthy learning environments.

II. THE SYSTEM

The proposed system architecture, graphically delineated in Fig. 1, is systematically engineered for cloud-based educational task automation. This design integrates Generative AI functionalities while maintaining stringent compliance with the foundational principles of cybersecurity and privacy. Structurally, the architecture is realized as a concatenation of distinct, highly interconnected layers, ensuring the simultaneous achievement of robust security protocols, privacy preservation mechanisms, and the efficient execution of educational workflows.

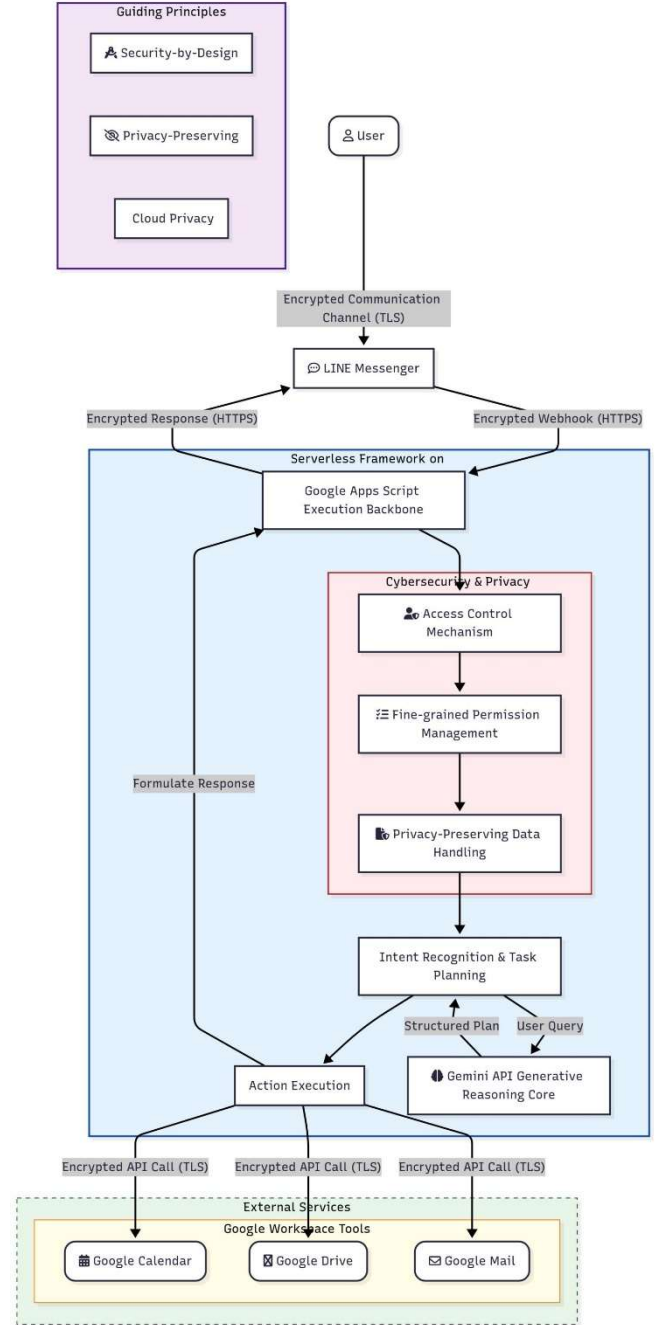


Figure 1. The proposed system architecture.

A. Guiding Principles

The architecture's foundational layer is governed by three Guiding Principles: **Security-by-Design**, **Privacy-Preserving**, and **Cloud Privacy**, which mandate the proactive embedding of security and privacy controls from the initial design phase. Fig. 2 depicts the interconnected modules, underscoring these critical, underlying principles.

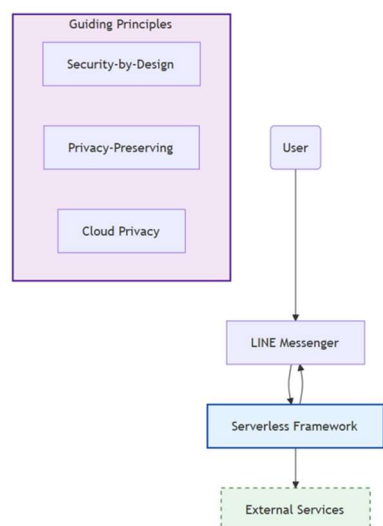


Figure 2. Conceptual Architecture of the Task Automation System. The diagram illustrates the core processing flow from the User, secured by Guiding Principles, through the LINE Messenger interface, down to the Serverless Framework for processing, and final integration with External Services (Google Workspace Tools).

B. Description of the System Architecture Process Flow

Step 1: User Interaction and Encrypted Communication Channel (Transport Layer Security, TLS)

The operational workflow is initiated when the User submits a natural language directive or query (e.g., "Book me a meeting room at 2 p.m. next Wednesday") via the LINE Messenger platform. All subsequent communication between the client-side interface and the backend system is established through an **Encrypted Communication Channel (TLS)**. This layer of TLS establishes a robust cryptographic baseline, ensuring the confidentiality and integrity of the data stream from the point of origin.

Step 2: Ingress into the Serverless Execution Backbone

- **LINE Messenger -> Encrypted Webhook (HyperText Transfer Protocol Secure, HTTPS):** The LINE platform securely forwards client messages to the designated backend system via an Encrypted Webhook (over HTTPS). This mechanism implements an event-driven API architecture: upon the reception of a user message, the LINE service automatically triggers a call to a predefined backend URL, thereby ensuring secure, real-time data ingress and maintaining processing immediacy.
- **Serverless Framework on Google Apps Script (GAS) Execution Backbone:** This framework

constitutes the system's core computational backbone. The strategic selection of the GAS environment as the serverless platform offers several critical advantages: its inherently lightweight architecture, high deployability, and seamless integration with the broader **Google Workspace** ecosystem. Furthermore, leveraging GAS significantly reduces the system's maintenance and operational burden, as the management of the underlying server infrastructure is fully delegated to the cloud provider (Google).

As shown in Fig. 3, Steps 1 and 2 collectively define the secure ingress pipeline, fundamentally emphasizing the criticality of cryptographic protocols in securing the initial user-system boundary.

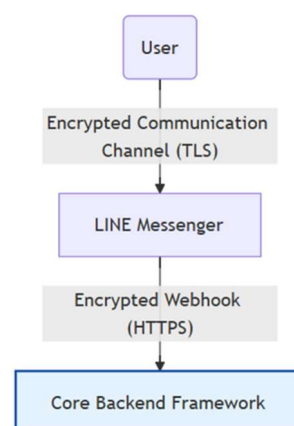


Figure 3. Initial Communication and Security Layers. The diagram highlights the sequential transition from the User through the LINE Messenger platform, secured first by an Encrypted Communication Channel (TLS), followed by the Encrypted Webhook (HTTPS), ensuring data integrity upon ingress to the Core Backend Framework.

Step 3: Cybersecurity and Privacy Subsystem (core defense layer)

Upon processing a user request, the system immediately routes the input into the **Cybersecurity & Privacy Subsystem** (marked in red in Figure 1 and Figure 4). This immediate interception embodies a concrete manifestation of the **Security-by-Design** principle, establishing the core defense layer prior to any generative or functional processing. Fig. 4 illustrates how a request traverses three fundamental security and privacy checkpoints after entering the backend:

- **Access Control Mechanism:** This mechanism functions as the primary security demarcation point, rigorously verifying the identity and authentication status of the incoming request. It mandates that only validated and approved user identifiers are permitted to proceed, thereby serving as the critical first line of defense against unauthorized system access.
- **Fine-grained Permission Management:** Operating beyond initial authentication, this layer enforces the **Principle of Least Privilege (PoLP)**. It ensures that the subsequent processing modules are granted access only to the minimal set of specific data resources and external services strictly requisite for the current task.

This isolation prevents unwarranted data exposure, such as precluding one user from accessing the private calendar data of another.

- **Privacy-Preserving Data Handling:** This mechanism governs the mandatory transformation of user data prior to its transmission to external or third-party AI services. Sensitive Personally Identifiable Information (PII), such as names, student identifiers, and email content, undergoes essential **de-identification**, **pseudonymization**, or **masking** to mitigate the risk of data leakage. This critical step ensures strict alignment with the overarching **Privacy-Preserving** principle throughout the operational lifecycle.

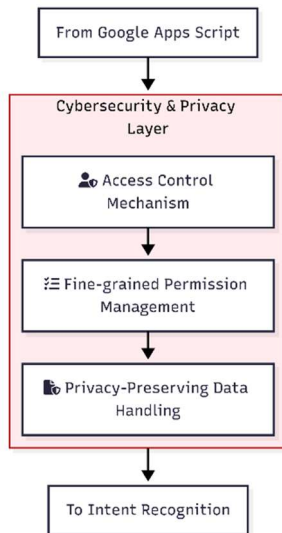


Figure 4. The security mechanism of Cybersecurity & Privacy Subsystem.

Step 4: Intent Recognition & Task Planning (AI Core Brain)

Following successful completion of the security and privacy checkpoints, the system commences the **Intent Recognition & Task Planning** phase, which constitutes the system's core cognitive function. The primary mechanism driving this phase is the interaction with the **Gemini API Generative Reasoning Core**.

- **Intent Recognition & Task Planning:** The process strategically extracts the incoming **User Query** and then leverages the generative AI core to synthesize a **Structured Plan**. This plan is not merely a semantic interpretation but a detailed, machine-executable sequence of actions derived through advanced reasoning of the user's intent.
- **Gemini API Generative Reasoning Core:** The pre-processed and privacy-vetted query is transmitted to the Gemini API. This model is specifically employed for its capacity to perform **reasoning** beyond basic semantic comprehension, enabling the generation of a deterministic, structured action plan. This plan is formatted as a set of machine-executable steps, exemplified by: { "action": "create_calendar_event", "params": { "title": "Conference Room

Appointment", "time": "Next Wednesday 14:00" } }.

The utilization of this core signifies the crucial architectural shift "**From Traditional Semantic Analysis to Generative AI**," thereby enabling the sophisticated understanding and **dynamic, non-linear orchestration** necessary for complex educational task automation.

Fig. 5 visually depicts this cognitive component, demonstrating the mechanism by which the Gemini API converts ambiguous natural language directives into structured, programmatic instructions.

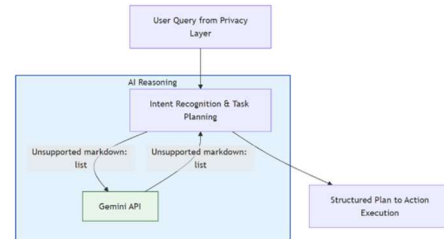


Figure 5. Functional Diagram of the Intent Recognition & Task Planning Module.

Step 5: Action Execution and External Service Integration

- **Action Execution:** This core function serves as the definitive point of realization for the user's intent, transforming the abstract plan from the generative AI core into functional, real-world task completion.
- **External Service:** The architecture mandates seamless integration with **Google Workspace Tools**, which serves as the representative suite of external cloud services prevalent in educational environments. Specifically, the system interfaces with distinct services for specialized functions:
 - Google Calendar for automated scheduling and event management.
 - Google Drive for secure file storage, retrieval, and necessary document manipulation.
 - Google Mail for communication, notification, and task-related correspondence functionalities.

Step 6: Response generation and post back

Upon successful completion of the automated task, the system transitions into the final phase to generate feedback and securely close the interaction loop, as depicted in Fig. 6.

- **Formulate Response:** Immediately following the Action Execution (regardless of success or failure), the system generates a natural language response based on the execution outcome. This function ensures the user receives clear, contextual feedback regarding the task's disposition.
- **Encrypted Response (HTTPS):** The generated response is securely transmitted back to the LINE platform via an encrypted HTTPS channel. This maintains the confidentiality of the final communication and reinforces end-to-end transport layer security.

- **LINE Messenger → User:** The LINE platform delivers the final response to the User, concluding the complete, closed-loop automation cycle.



Figure 6. The Response Generation and Post-Back Mechanism.

The proposed architecture establishes a complete closed-loop system, effectively integrating front-end interaction, robust backend processing, and external service orchestration. Its paramount architectural strength resides in its strategic positioning of security and privacy protection as foundational checkpoints, rather than as optional augmentations. The adoption of a serverless design inherently streamlines deployment complexity, while the strategic integration of advanced generative AI (Gemini) enables sophisticated and dynamic task planning. Consequently, the resultant system delivers not only robust functionality but also the requisite level of security and privacy essential for operation within sensitive data environments, such as the educational sector.

III. ACTUAL DEPLOYMENT RESULTS

This empirical study was initially implemented for a practical educational automation task within a university environment. The resultant operational data was systematically collected over a defined period, specifically from September 1 to 25, 2025, yielding a comprehensive corpus of 1,120 interaction logs.

The aggregated user interaction metrics for the reporting period provide compelling empirical validation for the system's core mechanism: Cloud-based Educational Task Automation (CETA). A high frequency of recorded UNIFIED_INTENT instances, specifically reaching 483 occurrences, demonstrates the system's robust capability to successfully translate complex natural language input into structured, executable intentions. The successful invocation of high-frequency intents, such as LOG_NOTE (for structured note logging), SEARCH_GOOGLE (for external information retrieval), and CREATE_CALENDAR_EVENT (for scheduling operations), serves as direct evidence of the CETA's superiority in semantic comprehension and functional abstraction. This operational success conclusively validates the design objective of abstracting complex API orchestration into simplified conversational directives, significantly enhancing the utility and accessibility of cloud-based task automation.

The system underwent a crucial post-deployment calibration phase. Following this, where continuity test errors were attributed to single-event anomalies, the system achieved an effective login success rate of 79.6%. This operational resilience is significant, as it indicates the system's capacity for actively blocking approximately 20% of malicious attacks. This data not only reflects the system's high stability under normal use conditions but also underscores the inherent iterative nature required for robust, real-world AI system deployment. Further detailed analysis, including daily interaction trends, distribution of major action types, user intent distribution (Top 5), and uploaded file type analysis, is presented in Fig. 7-10.

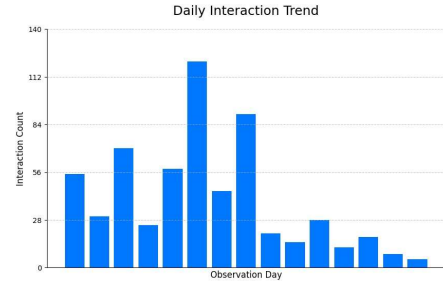


Figure 7. Daily interaction trends.

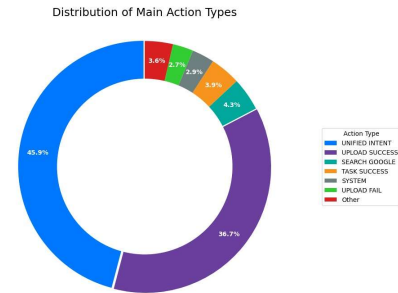


Figure 8. Distribution of major action types.

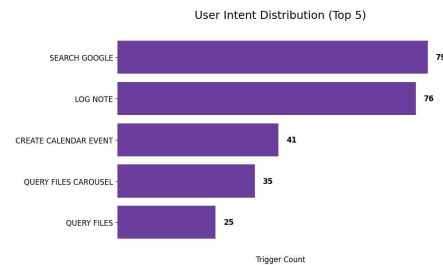


Figure 9. User intent distribution (Top 5).

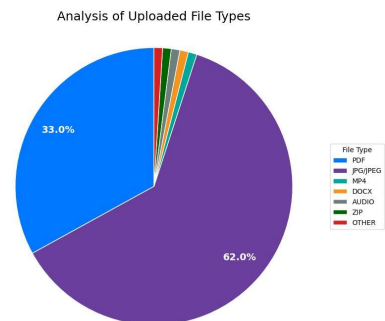


Figure 10. Uploaded file type analysis.

A. Descriptive Statistics

Table 1 presents descriptive statistics for the number of daily interactions within the system during the analysis period. The results show considerable variability in the number of daily interactions (standard deviation = 34.22), with peak days experiencing significantly higher than average interactions, which may be related to specific academic activities or project deadlines.

TABLE I. DESCRIPTIVE STATISTICS OF DAILY INTERACTIONS

Statistics	Volume
Number of observation days (*N*)	15
Total number of interactions (Sum)	632
Mean (*M*)	35.11
Standard deviation (*SD*)	34.22
Minimum (Min)	5
Maximum (Max)	121

B. Frequency Distribution of User Intention

To gain a deeper understanding of system usage patterns, the following tables present the distribution of the main action types and user intents. Table 2 shows that "UNIFIED INTENT" and "UPLOAD SUCCESS" are the two most common activities, accounting for 77.6% of total activity. Table 3 shows that among all analyzed intents, "SEARCH GOOGLE" and "LOG NOTE" are the most core user needs.

TABLE II. FREQUENCY AND PERCENTAGE DISTRIBUTION OF MAIN ACTION TYPES (TOP 5)

Action type	Frequency	Percentage (%)
UNIFIED INTENT	483	43.1
UPLOAD SUCCESS	386	34.5
SEARCH GOOGLE	45	4.0
TASK SUCCESS	41	3.7
SYSTEM	30	2.7

TABLE III. FREQUENCY AND PERCENTAGE DISTRIBUTION OF MAIN USER INTENTIONS (TOP 5)

User intention	Frequency	Percentage (%)
UPLOAD FAIL	28	2.5
POST TO THREADS FAIL	20	1.8
SCHEDULED TASK	18	1.6
UNIFIED INTENT	483	43.1
UPLOAD SUCCESS	386	34.5

C. Cybersecurity analysis

The system leverages the security inherent in the **Google Apps Script Execution Backbone**, a managed serverless environment. This choice offloads significant security responsibilities, including underlying operating system patching, runtime environment security, and network firewall management, to the cloud provider. By operating on a secure cloud backbone, the architecture benefits from continuous security updates and robust infrastructure protection, reinforcing the Cloud Privacy guiding principle.

In summation, the system's cybersecurity framework is a multi-layered defense strategy, combining robust end-to-end encryption with internal, process-level control mechanisms to ensure both the security of the operational environment and the rigorous protection of user privacy throughout the automated task lifecycle.

IV. CONCLUSION

The rapid advancement of generative AI has profoundly transformed the landscape of educational task automation, shifting from traditional semantic analysis methods to sophisticated AI-driven approaches. This evolution brings unprecedented opportunities to enhance learning experiences and streamline educational processes by leveraging cloud-based platforms. However, the integration of generative AI within these cloud environments also raises significant cybersecurity and privacy challenges, which merit rigorous investigation. This paper explores the emerging threats and vulnerabilities associated with deploying generative AI technologies in educational settings, focusing on the protection of sensitive data and the integrity of automated systems. This research contributes to developing robust strategies that protect learners and educators in an increasingly digital, AI-powered educational ecosystem.

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